

Drishti Obstacle Detection - Model Evaluation Report

Model Evaluation Report

Model: YOLOv11n | Drishti Obstacle Detection System

1. Metrics Summary

Metric	Value
mAP@50	0.9574
mAP@50-95	0.6946
Precision	0.9418
Recall	0.919
Accuracy	0.8123
F1 Score (Weighted)	0.811
Kappa Score	0.7805
MCC Score	0.7818
CV Mean Accuracy	0.9752
CV Mean F1	0.9035
ROC-AUC (motorcycle)	0.9945
Best F1 (motorcycle)	0.9091
ROC-AUC (person)	0.9879
Best F1 (person)	0.96
ROC-AUC (plant pot)	1.0
Best F1 (plant pot)	0.9961
ROC-AUC (pole)	0.9868
Best F1 (pole)	0.9497
ROC-AUC (pothole)	0.9971
Best F1 (pothole)	0.9901
ROC-AUC (tree)	0.999
Best F1 (tree)	0.9822
ROC-AUC (zebra cross)	0.9997
Best F1 (zebra cross)	0.9933

2. Overfitting Analysis

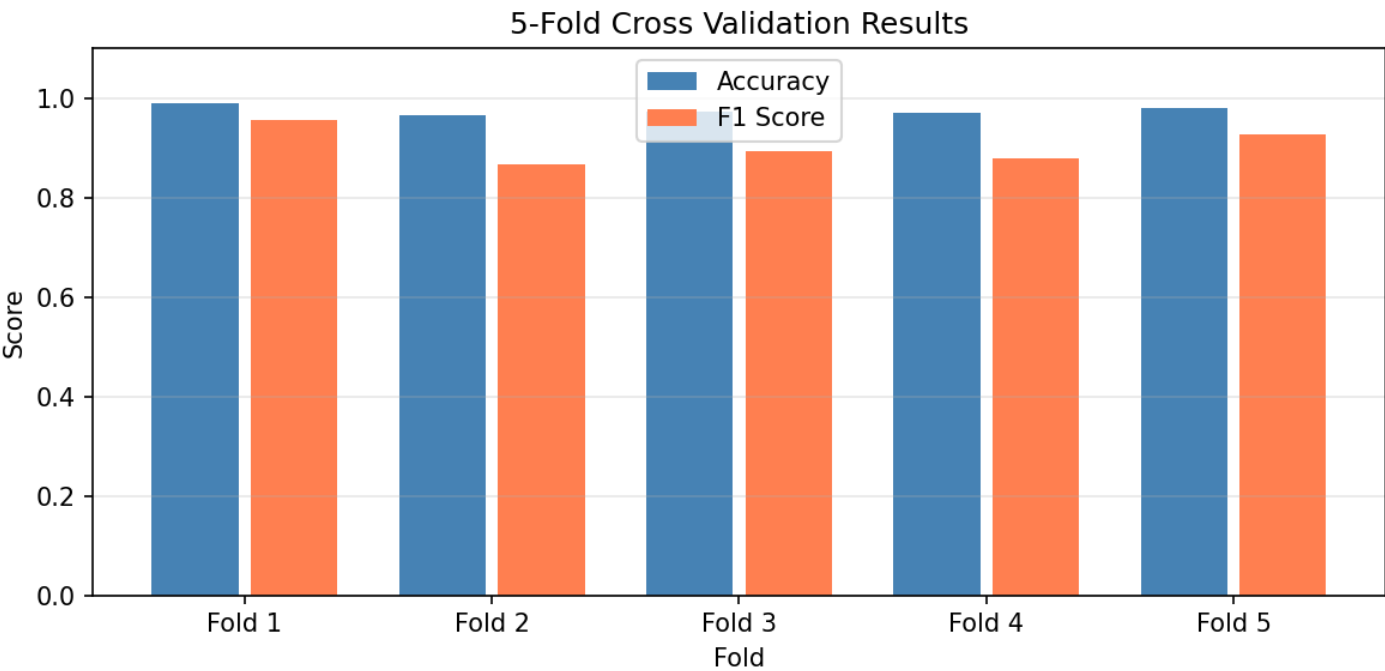
CV Mean Accuracy : 0.9752

Drishti Obstacle Detection - Model Evaluation Report

Test Accuracy : 0.8123

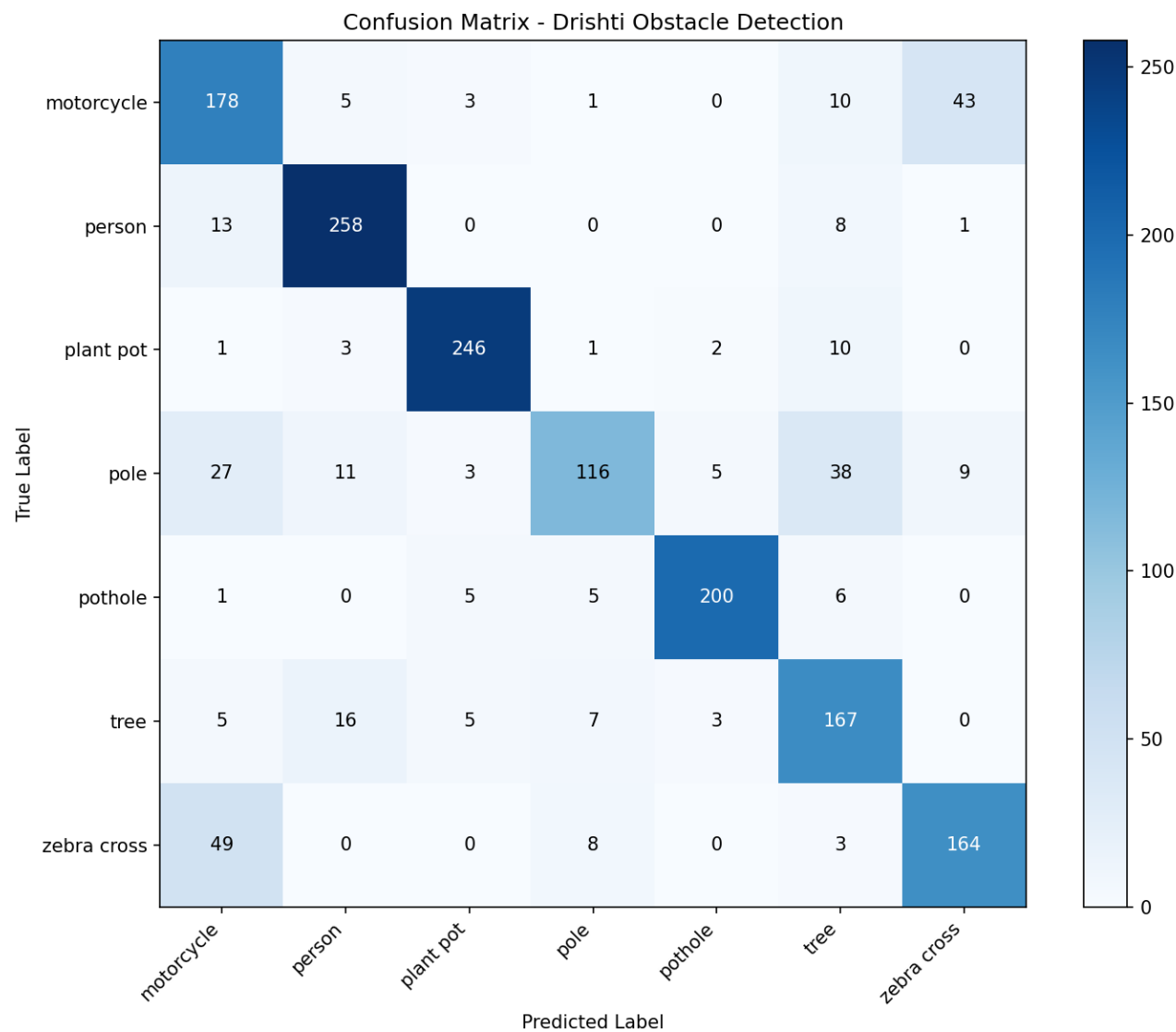
Difference : 0.1628 - Slight overfitting

3. K-Fold Cross Validation



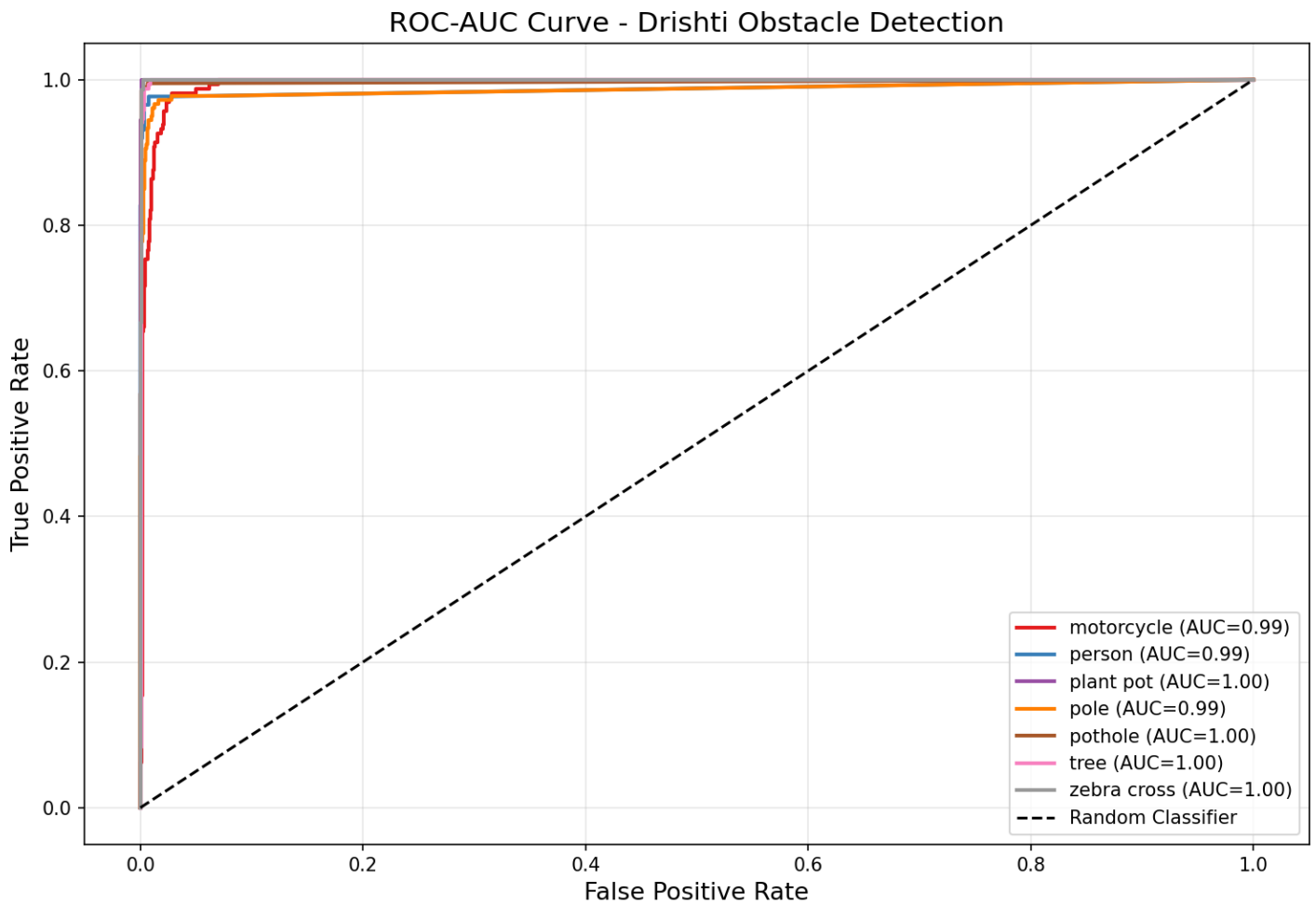
Drishti Obstacle Detection - Model Evaluation Report

4. Confusion Matrix



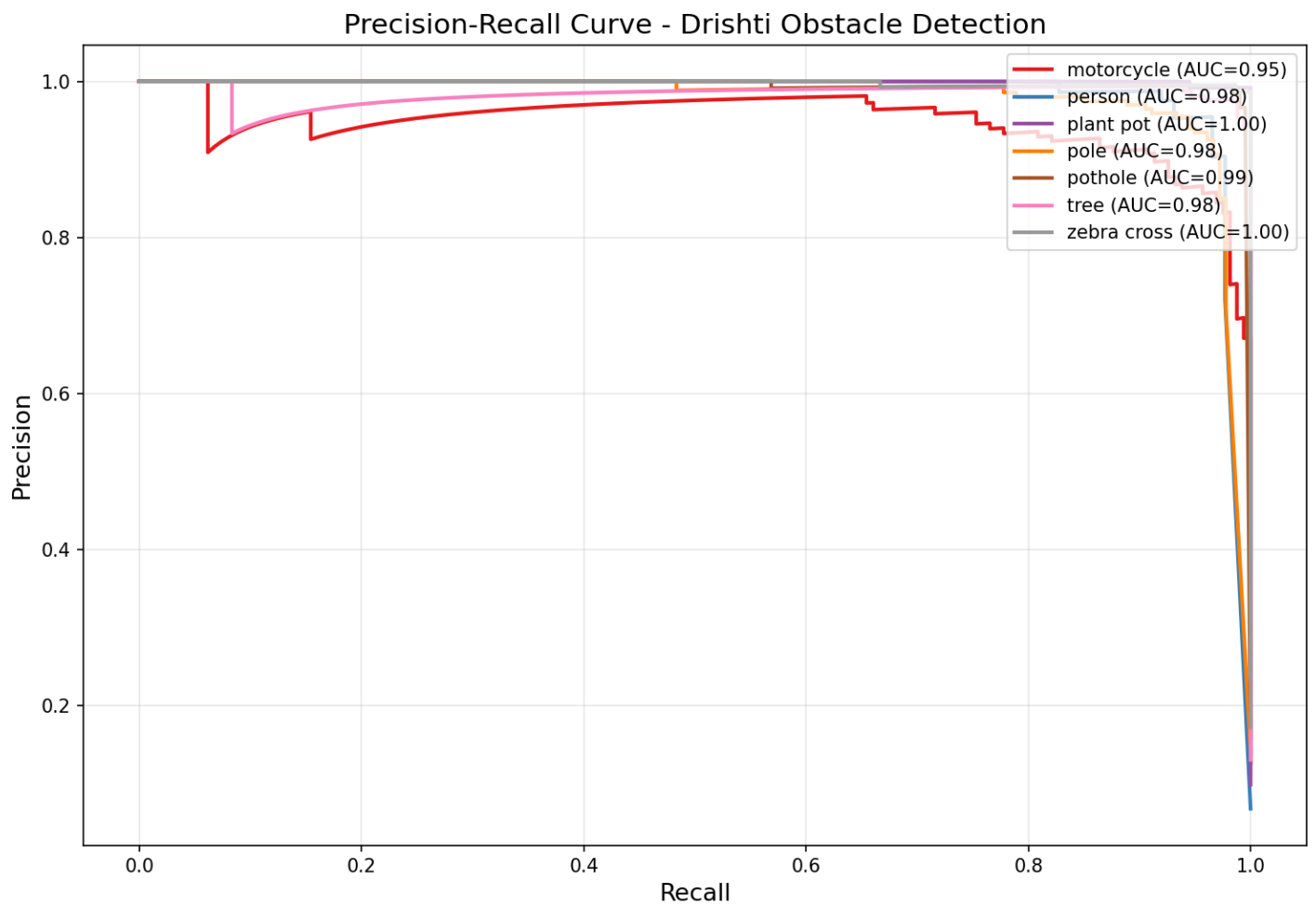
Drishti Obstacle Detection - Model Evaluation Report

5. ROC-AUC Curve



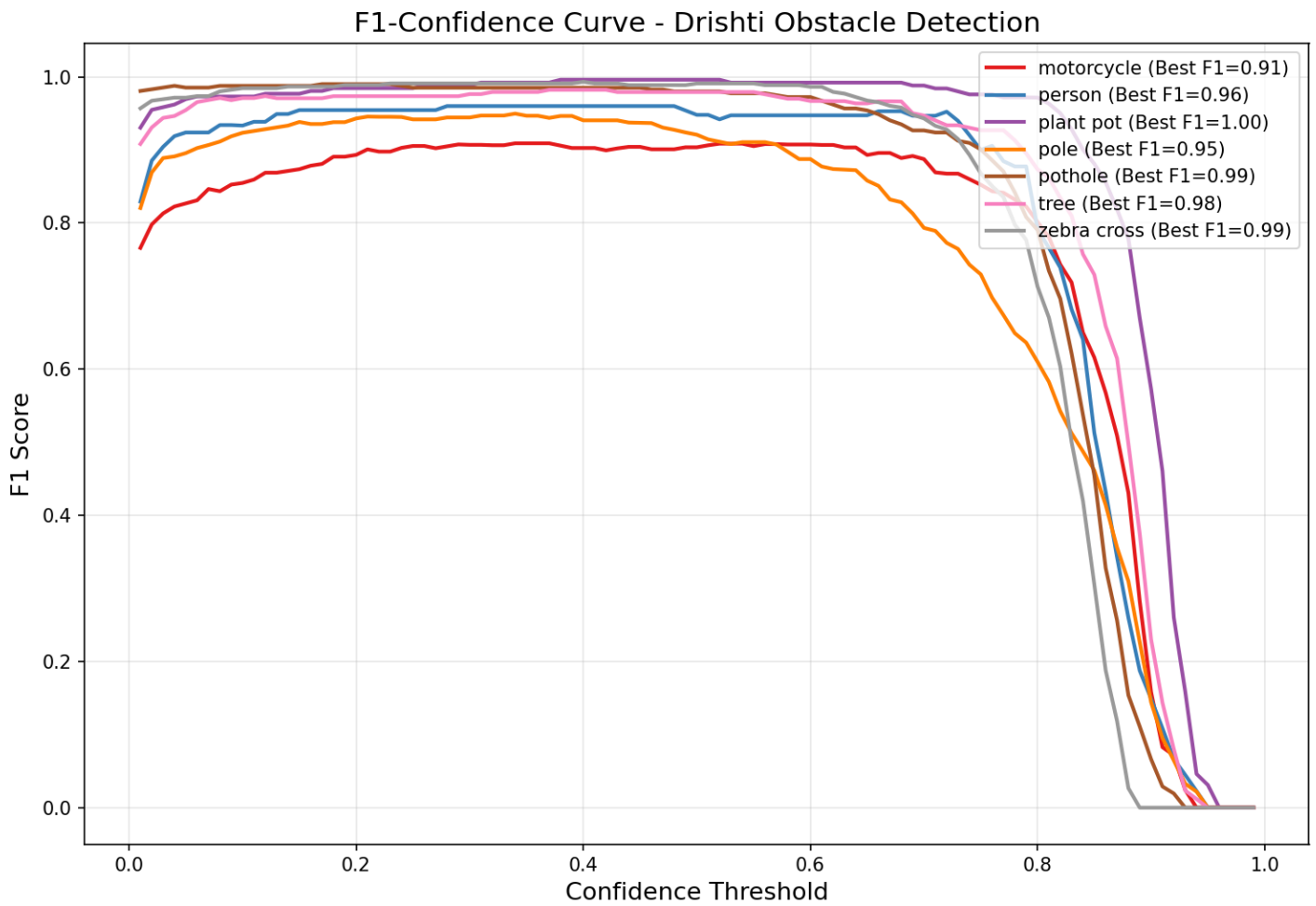
Drishti Obstacle Detection - Model Evaluation Report

6. Precision-Recall Curve



Drishti Obstacle Detection - Model Evaluation Report

7. F1-Confidence Curve



Drishti Obstacle Detection - Model Evaluation Report

8. Overfitting Check

